TY BSc Computer Science

Operating System Practical – Viva Questions

Quick Revision PDF • 30 Practical Slips

1. Basic Operating System

Q. What is an Operating System?

Ans: System software that manages hardware, software resources and provides services to programs.

Q. What is a process?

Ans: A program in execution.

Q. What is a system call?

Ans: An interface through which a program requests a service from the operating system.

Q. What is a shell?

Ans: A command interpreter that accepts user commands and executes them.

Q. What is a process ID (PID)?

Ans: A unique number assigned to a process by the operating system.

2. Process and System Calls

Q. What is fork()?

Ans: It creates a new child process from the calling process.

Q. What does fork() return?

Ans: 0 to the child, the child PID to the parent, and -1 on failure.

Q. What is execve()?

Ans: It replaces the current process image with a new program.

Q. What is wait()?

Ans: It makes the parent wait for a child process to finish.

Q. What are getpid() and getppid()?

Ans: getpid() returns the current process ID; getppid() returns the parent process ID.

Q. What is an orphan process?

Ans: A child process whose parent has terminated while the child is still running.

Q. What is nice()?

Ans: A system call used to change a process's scheduling niceness/priority value.

3. Disk Scheduling

Q. What is disk scheduling?

Ans: The method used by the OS to decide the order in which disk I/O requests are served.

Q. What is FCFS?

Ans: First Come First Serve: requests are served in their arrival order.

Q. What is SSTF?

Ans: Shortest Seek Time First: the request closest to the current head is served first.

Q. What is SCAN?

Ans: The disk head moves in one direction serving requests, reaches the end, then reverses direction.

Q. What is C-SCAN?

Ans: The head services requests in one direction; after reaching the end, it returns to the beginning and continues.

Q. What is LOOK?

Ans: Like SCAN, but the head reverses at the last pending request instead of going to the physical end.

Q. What is head movement?

Ans: The distance traveled by the disk head between serviced requests.

4. Page Replacement

Q. What is paging?

Ans: A memory-management technique that divides logical memory into pages and physical memory into frames.

Q. What is a page fault?

Ans: It occurs when a required page is not currently present in physical memory.

Q. What is FIFO page replacement?

Ans: It replaces the page that has been in memory for the longest time.

Q. What is LRU?

Ans: Least Recently Used: it replaces the page that has not been used for the longest time.

Q. What is Optimal page replacement?

Ans: It replaces the page whose next use is farthest in the future.

Q. What is MFU?

Ans: Most Frequently Used: it selects the page with the highest usage frequency according to the implemented policy.

Q. What is Belady's anomaly?

Ans: In FIFO, increasing the number of frames can sometimes increase the number of page faults.

5. Deadlock and Banker's Algorithm

Q. What is deadlock?

Ans: A situation where processes wait indefinitely for resources held by each other.

Q. What is Banker's Algorithm?

Ans: A deadlock-avoidance algorithm that grants resources only if the resulting state remains safe.

Q. What is a safe state?

Ans: A state in which there is at least one sequence allowing all processes to complete.

Q. What is the Need matrix?

Ans: Need = Max - Allocation.

Q. What is a resource request?

Ans: The additional resources currently requested by a process.

6. File Allocation

Q. What is file allocation?

Ans: The method used by the OS to allocate disk blocks to files.

Q. What is contiguous/sequential allocation?

Ans: A file is stored in consecutive disk blocks.

Q. What is linked allocation?

Ans: Each file block points to the next block, forming a linked chain.

Q. What is indexed allocation?

Ans: An index block stores pointers to the data blocks of a file.

Q. What is a bit vector?

Ans: A bit array in which each bit represents the allocation/free status of a disk block.

Q. What is fragmentation?

Ans: Wasted or unusable memory/disk space caused by the way storage is allocated.

7. Important C / Linux Commands for Practical

Q. How do you compile a C program using GCC?

Ans: gcc program.c -o program

Q. How do you run the compiled program?

Ans: ./program

Q. Why is #include used?

Ans: It provides declarations for many POSIX/Linux system calls such as fork(), exec(), getpid() and sleep().

Q. Why is #include used?

Ans: It provides functions/macros related to waiting for child processes, such as wait().

Q. Why is #include used?

Ans: It provides functions such as exit(), atoi(), rand() and abs().

8. Very Important One-Line Questions

Q. Which algorithm gives the theoretical minimum page faults?

Ans: Optimal page replacement.

Q. Which disk algorithm can cause starvation of far-away requests?

Ans: SSTF can cause starvation.

Q. What does 0/1 commonly represent in a bit vector?

Ans: It depends on the program convention; in our file-allocation programs, 0 = free and 1 = allocated.

Q. What is the main aim of disk scheduling?

Ans: To reduce disk head movement/seek time and improve I/O performance.

Q. What is the difference between parent and child process?

Ans: The child is created by the parent using fork(); both have separate process IDs.

Exam Tip: First focus on running the code correctly. Remember the input format, main logic, and expected output for each practical.